Situation-Aware Player Similarity via Trait–State Decomposition on Heterogeneous Graphs

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Abstract

We study substitute-oriented soccer player retrieval: given a query player, find those who would make similar on-ball decisions under the same tactical state. We model each possession as a heterogeneous graph over events and 360 freeze-frame players, learn contextual state embeddings with a GATv2 encoder, extract global trait vectors via attention pooling, and condition action prediction through FiLM modulation. A systematic ablation across 11 pipeline variants examines eight modular components, with three heuristic baselines used as supplementary reference points.

Four main observations emerge from the tested variants. First, in this study, position ablation combined with split-context edges and stronger uniformity settings yields the most stable non-collapsed embeddings and the strongest cross-context consistency among the configurations evaluated. Second, player-aware sampling appears to optimize player re-identification more than cross-player substitute discovery. Third, a coarse empirical behavioral (EB) metric indicates that the acts-in-dropout family produces stronger observed behavioral matching than random same-position peers, although the EB protocol is too coarse to cleanly isolate individual component effects. Fourth, under the chosen weighted composite, `acts_in_dropout_pos_gu` attains the highest overall score (0.867), reflecting the strongest overall balance across behavioral fidelity, cross-context robustness, and external validation rather than dominance on every individual metric.

1 Introduction

The goal is to retrieve soccer players who function as plausible *substitutes* for a query player: not merely similar in season-level production, but similar in how they would decide under the same tactical situation. Aggregate statistics confound player behavior with team quality, role assignment, and tactical scheme. A fullback on a dominant side and one on a transitional side may post very different numbers even if their decision-making under matched states is identical.

We formalize this with a behavioral discrepancy:

$$\Delta(p, q) = \mathbb{E}_{s\sim\mathcal{D}_S}\Big[\mathrm{JS}\big(P_\theta(\cdot \mid s, z_p),\ P_\theta(\cdot \mid s, z_q)\big)\Big],$$

where s is a masked pre-action game state, $z_p \in \mathbb{R}^d$ is a learned trait embedding, and P_θ is the conditional action policy. The practical goal is to learn z_p so that cosine proximity approximates low $\Delta(p, q)$.

Section 3 describes data and state representation. Section 4 presents the model architecture. Sections 5–6 detail the training objective and the eight modular components studied. Sections 7–8 present evaluation and results, followed by model selection (Section 9), discussion (Section 10), and conclusion (Section 11).

2 Related Work

Conventional player comparison relies on aggregated statistics or composite ratings. Decroos et al. [3] (VAEP) and Fernández et al. [4] value individual actions and possessions but do not yield continuous player embeddings suitable for retrieval. GNNs have been applied to team-sport analysis: Yeh et al. [11] model basketball trajectories as spatio-temporal graphs with graph neural networks and variational RNNs, enabling counterfactual trajectory prediction. Our architecture draws on GATv2 [2], FiLM [8], InfoNCE [7], focal loss [6], and alignment and uniformity analysis on the hypersphere [10]. Our contribution is integrative: the individual components are established, but their combination for behavioral player similarity, and the systematic ablation revealing which interactions matter, is novel in this domain.

3 Data and State Representation

The system uses StatsBomb 360 open data [9], which augments standard event data with freeze-frame snapshots of all visible players at each on-ball action. The working corpus contains approximately 51,000 possessions and 737,000 events across 323 matches from 7 competition-seasons with 360 coverage. Players with fewer than 50 possessions are excluded, yielding 1,633 embedded players (311 embedded players appear in $\geq$2 competitions).

Each event is encoded as a 126-dimensional vector mixing categorical fields (event type, play pattern, position, body part, action/outcome descriptors), continuous fields (pitch coordinates, displacement, pass geometry, pressure), and contextual fields (pitch zone, match period, spatial-360 summaries from freeze frames). The spatial-360 block is zeroed in the graph representation so the GNN must learn spatial context from explicit player nodes.

Events are grouped by `possession_number` into possession sequences with per-event timestamps. Possessions with fewer than 2 or more than 200 events are discarded. Possession-level labels (ends-in-shot, ends-in-goal, total xG) provide a secondary supervisory signal.

4 Architecture

The architecture decomposes each prediction into a *state pathway* (what the current situation affords) and a *trait pathway* (how a specific player tends to act). Figure 1 shows the graph topology; Figure 2 illustrates the full pipeline.

Heterogeneous graph. Each possession is encoded as a heterogeneous graph with two node types: **event nodes** (masked 126-D features) and **player nodes** ([position_idx, team_flag, dx, dy], covering both the on-ball actor and off-ball players from the 360 freeze frame). Edges encode temporal adjacency (with normalized time-delta), actor–event links (`acts_in` / `performed_by`), and context edges connecting off-ball players to their focal event. When split-context mode is active (Section 6), context edges separate into teammate and opponent relations.

GNN encoder. A two-layer heterogeneous GATv2 [2] encoder (`HeteroConv` [5]) processes the graph: Layer 1 maps 64-D inputs to 512-D (4 heads $\times$ 128-D); Layer 2 projects back to 64-D (single head), with learned skip connections.

Strict masking. To prevent label leakage, we zero out event type (the prediction target), end location, displacement, position, body part, action/outcome descriptors, and action-specific scalars. Unmasked features preserve ball location, play pattern, pressure, pitch zone, and match period.

Global trait extraction. A player appearing across many possessions accumulates per-possession embeddings, aggregated via learned attention pooling [1]:

$$z_p = \sum_i \alpha_{p,i}\, h_{p,i}^{\text{player}}, \qquad \alpha_{p,i} = \text{softmax}_i(g(h_{p,i}^{\text{player}})),$$

where g is a two-layer scoring network.

FiLM conditioning. Prediction uses Feature-wise Linear Modulation [8]:

$$\widetilde{h}^{\text{event}}(p) = \big(1 + \gamma(c)\big) \odot h^{\text{event}} + \beta(c),$$

where $c = [z_p;\ r_p]$ and r_p is an optional 16-D positional embedding. Prediction heads output logits for action type (14-way), angle bin (9-way), and length bin (5-way).

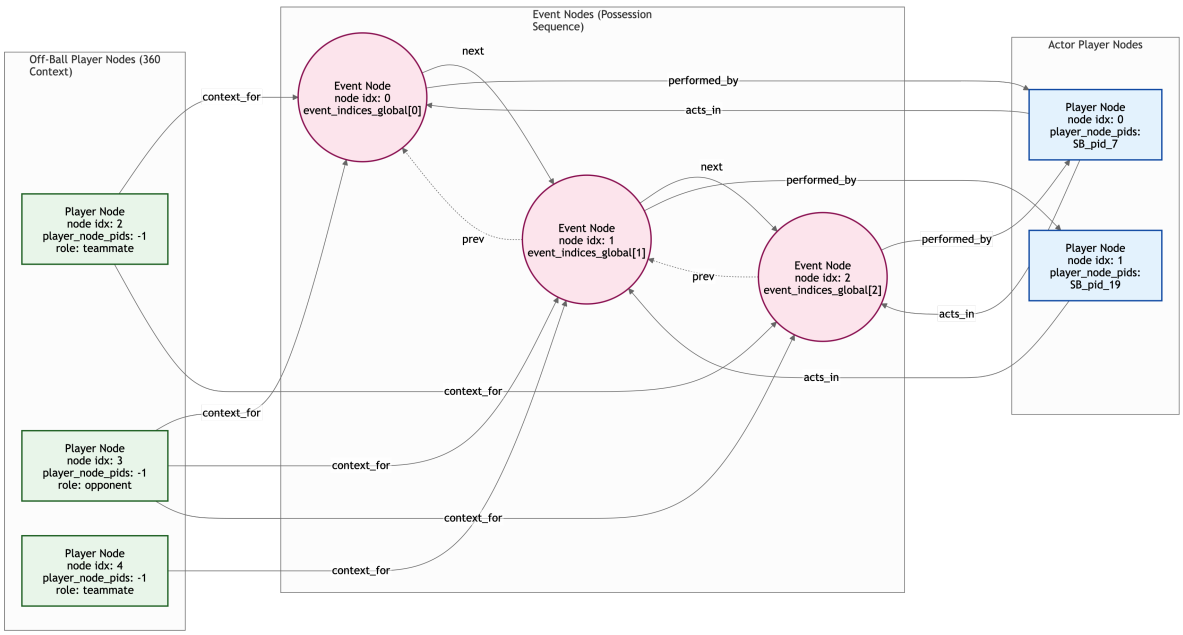


Figure 1: Heterogeneous graph topology for a single possession. Pink circles are event nodes linked by temporal edges; blue rectangles are actor player nodes with real player IDs connected via `acts_in`/`performed_by`; green rectangles are off-ball player nodes from 360 freeze frames connected via `context_for` edges.

5 Training Objective

The total loss combines supervised prediction with embedding-geometry regularization:

$$\mathcal{L} = \mathcal{L}_{\text{action}} + \lambda_{\text{out}}\,\mathcal{L}_{\text{outcome}} + \lambda_{\text{con}}\,\mathcal{L}_{\text{contrast}} + \lambda_{\text{uni}}\,\mathcal{L}_{\text{uniformity}} + \lambda_{\text{align}}\,\mathcal{L}_{\text{align}} + \lambda_{\text{pos}}\,\mathcal{L}_{\text{pos}}.$$

$\mathcal{L}_{\text{action}}$ uses focal loss [6] over action type, angle, and length bins with class-rebalancing weights. $\mathcal{L}_{\text{outcome}}$ is BCE for shot/goal prediction from h^{event} alone. $\mathcal{L}_{\text{contrast}}$ is InfoNCE [7] on actor embeddings with same-player positives and same-position-group hard negatives. $\mathcal{L}_{\text{uniformity}}$ is the Gaussian-potential loss [10] on L2-normalized z_p, optionally with group-weighted repulsion. $\mathcal{L}_{\text{align}}$ provides cross-batch alignment via EMA memory bank. $\mathcal{L}_{\text{pos}}$ predicts coarse position group from z_p. Default weights: $\lambda_{\text{out}} = 0.5$, $\lambda_{\text{con}} = 0.5$, $\lambda_{\text{uni}} = 0.3$, others inactive. Training uses Adam ($\text{lr} = 10^{-3}$, weight decay 10^{-5}), gradient clipping at 1.0, batch size 96, 70/15/15 match-level splits, and ReduceLROnPlateau scheduling.

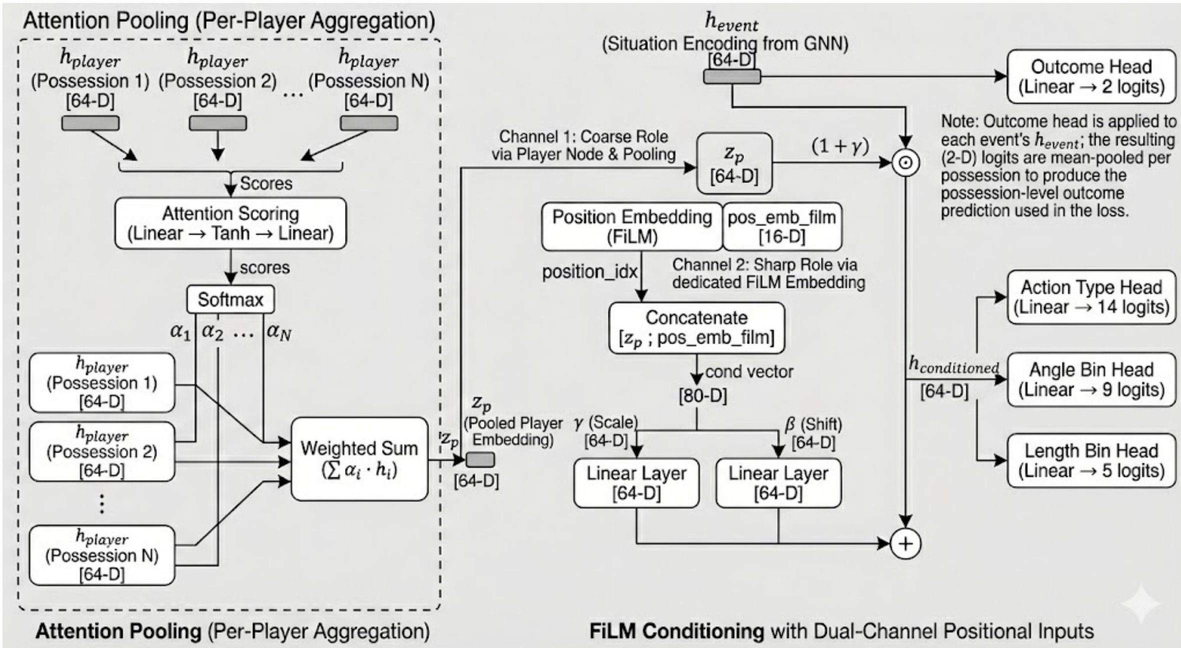


Figure 2: Full model architecture. *Left*: attention pooling aggregates per-possession player embeddings into a global trait z_p. *Right*: FiLM conditioning modulates the GNN's situation encoding with player-specific scale and shift. The outcome head operates on h^{event} alone, ensuring outcome prediction is player-independent.

6 Model Variation Components

We study eight modular components across 11 pipeline configurations (Table 1). Appendix E provides a complete hyperparameter specification for every variant.

C1. Split-context edges: separate teammate vs. opponent context relations.

C2. Position ablation: remove all explicit position information from the learned representation; FiLM receives z_p alone (64-D).

C3. Player-aware sampling: batch structure of K=16 players × M=6 possessions, with tuned contrastive preset.

C4. Uniformity pressure (t, λ_{uni}): Gaussian-potential sensitivity and weight; position-ablated variants use elevated settings (t=4.0, λ_{uni}=1.0).

C5. EMA alignment: cross-batch alignment loss with momentum memory bank (decay 0.999).

C6. Acts-in edge dropout: drops all actor→event edges with p=0.3 during training, forcing FiLM and z_p to carry more of the behavioral prediction burden.

C7. Position-group prediction (λ_{pos}): auxiliary loss predicting coarse position from z_p, reintroducing positional structure as a soft target.

C8. Group-weighted uniformity: 3× repulsive force for same-position-group pairs, sharpening within-group discrimination.

7 Evaluation Protocol

Each model is evaluated across 14 complementary metrics spanning six categories (see Appendix A for a detailed description of the evaluation protocols and methodologies):

Table 1: Pipeline variants and active components. Blank = default/inactive.

Model	Split Ctx	Pos Abl	Player Samp	t	λ uni	EMA	λ align	Acts-In Drop	λ pos	Group Wt
baseline				2	.3					1
player_samp			✓	2	.5					1
split_ctx	✓			2	.3					1
split_ctx_ps	✓		✓	2	.5					1
pos_ablated		✓		2	.3					1
pos_abl_split_ctx	✓	✓		4	1.0					1
pos_abl_split_ctx_v2	✓	✓		4	.7					1
pos_abl_split_ctx_ema	✓	✓		4	1.0	✓	.3			1
pos_abl_split_ctx_ema_v2	✓	✓		4	1.0	✓	.1			1
acts_in_dropout	✓	✓		4	1.0			.3		1
acts_in_dropout_pos_gu	✓	✓		4	1.0			.3	.3	3

Model-based behavioral fidelity (wt. 0.28): Spearman ρ between embedding cosine distance and JS divergence of predicted action distributions (within 8 position×gender groups); absolute top-k JS of the 10 closest neighbors; substitute quality ratio.

Empirical behavioral fidelity (wt. 0.09): non-self-referential check using observed actions bucketed by coarse game state (pitch third × under-pressure = 6 buckets, 5 action categories). EB Spearman ρ and EB substitute ratio.

Cross-context robustness (wt. 0.23): for 311 multi-competition players, competition-split self-consistency (mean rank and Hit@10).

Pseudo-ground-truth retrieval (wt. 0.12): a 15-pair heuristic probe set assembled from public analytics articles and media comparisons (see Appendix B). This is used only as a directional sanity check and should not be interpreted as expert-validated ground truth.

External validation (wt. 0.20): qualitative head-coach scoring (6 queries scored 1–5 by an LLM evaluator; the evaluator used was Claude Opus 4.6) and FIFA attribute comparison (16 non-GK pairs).

Supplementary (wt. 0.08): random-half self-consistency and held-out macro-F1.

Three heuristic baselines calibrate expectations: mean features, action-type histogram, and FIFA attributes.

8 Results

8.1 Held-out predictive validity

Table 2 reports held-out test performance for `acts_in_dropout_pos_gu`. Action-type accuracy reaches 0.839 (macro F1 0.713); outcome AUC exceeds 0.97. The F1 spread across non-collapsed models is small (0.050), with position-ablated variants paying a modest penalty for shifting capacity toward embedding geometry.

8.2 Retrieval and self-consistency

Table 3 presents pseudo-ground-truth retrieval and cross-context self-consistency side by side. The reported pseudo-GT mean ranks for the top models are relatively close, so they should not be interpreted as a clean statistical separation on their own. Player-sampling variants have conspicuously high median GT ranks (71–73) and the worst competition-split MR (>89), reflecting retrieval that does not generalize well across contexts. The `pos_abl_split_ctx` family is strongest on competition-split consistency (MR 67.3–67.7), with `acts_in_dropout` achieving the best competition-split Hit@10

Table 2: Held-out test metrics (`acts_in_dropout_pos_gu`).

Task	**Metric**	**Value**
Action Type (14-way)	Accuracy / Macro F1	0.839 / 0.713
Angle Bin (9-way)	Accuracy / Macro F1	0.674 / 0.609
Length Bin (5-way)	Accuracy / Macro F1	0.740 / 0.657
Outcome: Shot	AUC-ROC	0.975
Outcome: Goal	AUC-ROC	0.988

(0.351).

Table 3: Pseudo-GT retrieval (lower MR better; higher Hit better) and cross-context self-consistency.

	Pseudo-GT (15 pairs)				**Competition Split (311 players)**		
Model	MR	Med	H@10	H@50	Cos	MR	H@10
split_ctx	**56.1**	40	.233	.600	.859	83.9	.215
pos_abl_split_ctx_ema	62.6	37	.233	**.700**	.769	67.7	.318
pos_abl_split_ctx	63.1	39	**.267**	.633	.766	**67.3**	.346
acts_in_dropout_pos_gu	66.7	41	.233	.567	.746	74.0	.328
acts_in_dropout	68.2	32	**.267**	.633	.763	68.9	**.351**
baseline	67.6	**30**	.233	.667	.885	81.6	.238
player_samp	70.2	71	.200	.333	.569	93.5	.264
split_ctx_ps	71.4	73	.167	.333	.573	89.3	.262
pos_abl_split_ctx_v2	198.8	161	.067	.200	.325	200.8	.076

8.3 Behavioral fidelity: model-based and empirical

Table 4 consolidates model-based and empirical behavioral metrics. The highest model-based ρ (`pos_ablated`, 0.974) is partly consistent with embedding compression (mean cosine distance 0.085); among retrieval-viable models, `pos_abl_split_ctx_ema` leads (0.944). The EB rankings differ from the model-based rankings, suggesting that the EB view is not redundant with the model's own internal notion of similarity. `acts_in_dropout` leads EB substitute ratio (1.953), with `acts_in_dropout_pos_gu` second (1.745). Conversely, `pos_ablated`'s inflated model-based ρ does not carry over to EB ρ (0.196), indicating that rank-based compression effects are metric-specific.

8.4 External validation

The FIFA heuristic baseline leads FIFA-specific metrics because it directly optimizes that attribute space. Among the learned models, `acts_in_dropout_pos_gu` achieves the strongest external-validation profile shown here, with the best learned-model FIFA agreement (7.7) and the highest qualitative score (3.5/5). These results should still be interpreted cautiously: the qualitative judgments come from an LLM evaluator (Claude Opus 4.6), and the FIFA comparison is small (16 non-GK pairs). Accordingly, this section is best read as supportive evidence rather than definitive proof of tactical interchangeability.

8.5 Embedding structure and counterfactual validation

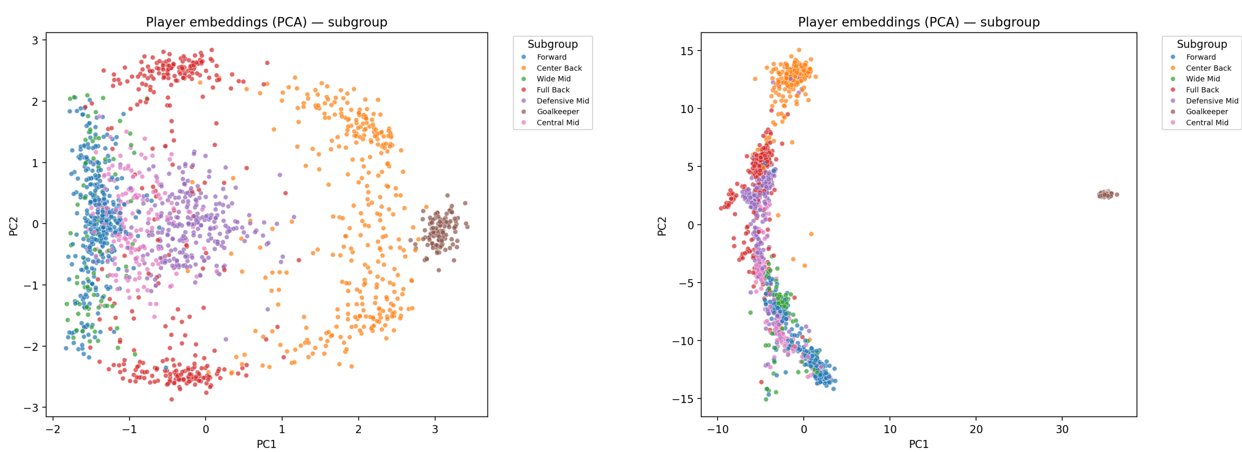
Trait-swapping validation confirms that the model's predictions change when different z_p are substituted into the same encoded state. A caveat is that, because actor–event edges are present at inference, some actor identity can leak into h^{event} before trait substitution. This therefore validates

Table 4: Behavioral fidelity. Model-based metrics use the model's own action predictions; EB metrics use observed player actions. Higher ρ and ratio are better; lower JS is better.

	Model-Based			**Empirical Behavioral**	
Model	ρ	Top-k JS	Ratio	EB ρ	EB Ratio
pos_ablated	**.974**	.00111	8.59	.196	1.66
pos_abl_split_ctx_ema	.944	.00104	7.95	.207	1.72
pos_abl_split_ctx	.929	.00141	6.63	.207	1.62
acts_in_dropout	.928	**.00103**	7.75	.207	**1.95**
acts_in_dropout_pos_gu	.922	.00105	7.62	.195	1.75
player_samp	.788	.00112	11.91	.223	1.56
split_ctx_ps	.778	**.00103**	**13.11**	**.230**	1.70
baseline	.744	.00330	3.84	.188	1.32
split_ctx	.710	.00461	3.13	.189	1.25

Table 5: External validation. FIFA: 16 non-GK query-neighbor pairs. Qualitative: 6 queries scored 1–5 by Claude Opus 4.6 (see Appendix D).

Model	FIFA M-6	FIFA Pos%	Qual. Total	Qual. Avg
acts_in_dropout_pos_gu	**7.7**	50	**21**/30	**3.5**
acts_in_dropout	7.9	50	18/30	3.0
pos_abl_split_ctx_ema	9.1	56	18/30	3.0
pos_abl_split_ctx	9.0	50	17/30	2.8
baseline	8.6	44	13/30	2.2
split_ctx	8.5	56	11/30	1.8
h_fifa_attributes	**5.0**	**75**	—	—



(a) Selected model embedding PCA: position subgroups remain visually separated despite removing positional input from FiLM.

(b) Baseline embedding PCA: position subgroup clustering is also visible, with tighter concentration in some groups.

Figure 3: PCA projection of global player embeddings by position subgroup.

FiLM sensitivity rather than a perfectly clean counterfactual intervention.

8.6 Qualitative neighbor inspection

Table 6 reports top-5 neighbors for selected query players, providing face-validity checks. Ronaldo retrieves clinical center forwards; Messi retrieves creative playmaker-forwards; De Bruyne retrieves progressive midfield creators; Bronze retrieves elite attacking right-backs from women's football. Gender-aware retrieval restricts Bronze's gallery to female players, yet her neighbors are stylistically meaningful. Face-validity for famous players is necessary but not sufficient; evaluating lesser-known or emerging players would require domain-expert judgments.

Table 6: Top-5 nearest neighbors (`acts_in_dropout_pos_gu`).

#	Query	Top-5 Neighbors	Cos
Ronaldo	Cristiano Ronaldo (CF)	Morata	.954
		Lautaro Martínez	.949
		Memphis Depay	.945
		V. Boniface	.940
		Kane	.935
Messi	Lionel Messi (RCF)	Griezmann	.939
		Shaqiri	.915
		Arda Güler	.907
		Kramarić	.893
		Mertens	.891
KDB	Kevin De Bruyne (CAM)	A. Ramsey	.911
		Zieliński	.901
		Bruno Fernandes	.893
		Griezmann	.889
		Miranchuk	.885
Bronze	Lucy Bronze (RB)	Lynn Wilms	.973
		Giulia Gwinn	.966
		Emily Fox	.965
		O. Hernández	.963
		E. Carpenter	.956

9 Weighted Model Selection

No single model dominates all metrics. To integrate trade-offs, we use a weighted composite with min-max normalization across the 11 GNN models:

$$S_m = \frac{\sum_i w_i \cdot \mathrm{norm}_{m,i}}{\sum_i w_i \cdot \mathbf{1}[\text{available}]}.$$

Weight justification and the full raw-metric table appear in Appendix C.

Under this weighting scheme, `acts_in_dropout_pos_gu` achieves the highest composite score (0.867), followed very closely by `acts_in_dropout` (0.864). Across four sensitivity analyses (balanced, behavioral-heavy, retrieval-heavy, and external-heavy), the two acts-in-dropout variants remain the top two models, suggesting that the high-level selection is moderately robust to reasonable weight changes.

Variant	#1	#2	#3
Balanced (default)	**Selected model** (.867)	Acts-in dropout (.864)	EMA-v2 (.860)
Behavioral-heavy	Acts-in dropout (.862)	**Selected model** (.857)	`pos_ablated` (.850)
Retrieval-heavy	Acts-in dropout (.877)	**Selected model** (.874)	EMA (.855)
External-heavy	**Selected model** (.878)	Acts-in dropout (.859)	`pos_ablated` (.834)

Caveats. The composite score is a model-selection aid, not a statistical proof of superiority. It depends on manually chosen weights, partial metric availability for some models, and evaluation protocols of uneven credibility. For example, `pos_ablated` is missing qualitative data, while `pos_abl_split_ctx_ema_v2` is missing both qualitative and EB results. Accordingly, the composite ranking should be interpreted as a transparent aggregation of available evidence rather than a definitive ordering.

10 Discussion

Progressive design insights. The 11 pipelines reflect a three-stage exploration. *Stage 1 (baselines)*: `split_ctx` achieves the best pseudo-GT mean rank (56.1) but weak behavioral fidelity and poor qualitative results, suggesting that positional structure remains too dominant in the embedding. *Stage 2 (position ablation)*: removing explicit positional input pushes the representation toward behavior, and the `pos_abl_split_ctx` family achieves the strongest competition-split consistency among the tested variants. In the present sweep, lowering the stronger ablation setting from $\lambda_{\text{uni}} = 1.0$ to 0.7 is associated with severe degradation, but this should be interpreted as evidence about the tested configuration grid rather than a universal threshold. *Stage 3 (acts-in dropout)*: dropping actor edges during training further increases reliance on the trait pathway. The EB analysis supports the acts-in-dropout family as the strongest observed-behavior match among the evaluated learned models, with group-weighted uniformity contributing more to qualitative coherence than to clear EB separation.

Key tensions. (1) *Player sampling vs. substitute discovery*: `player_samp` performs very strongly on self-consistency but poorly on cross-context substitute retrieval, indicating that "same player = close" is not the same objective as "different players with similar function = close". (2) *Behavioral fidelity vs. retrieval*: `pos_ablated` attains the highest model-based ρ but only middling retrieval quality, suggesting that compressed distance structure can inflate rank-based agreement. (3) *Pseudo-GT mean rank vs. qualitative quality*: `split_ctx` has the best pseudo-GT mean rank yet the weakest qualitative score among the displayed learned models, showing that these evaluation views are not interchangeable.

Role adjacency vs. tactical equivalence. The model captures similarity in on-ball decision distributions, but this does not always imply tactical interchangeability. Rúben Dias for Van Dijk is structurally coherent: both are elite ball-playing centre-backs who organize the line, step out, and distribute progressively, even though Dias is more aggressive and front-footed while Van Dijk favours calmer, longer distribution. Bruno Fernandes for De Bruyne is more role-adjacent than fully equivalent: both are creative midfielders who select similar action types (through-balls, crosses, shots) in advanced zones, but De Bruyne operates as a tempo-setting, half-space progression specialist within a controlled positional structure, whereas Bruno is more direct and transition-heavy. A 14-way action-type prediction cannot resolve such mechanistic distinctions; these would require modelling passing rhythm, off-ball positioning, and risk calibration that the current feature set does not capture. This suggests the model is better at capturing *what* decisions a player tends to make than fully capturing *how* those decisions are executed, which matters for tactical scouting.

Limitations.

- **Evaluation credibility**: external validation relies on LLM-compiled pairs, LLM-scored judgments, and a small FIFA comparison (16 pairs). Human expert evaluation would substantially strengthen the evidence.
- **Partially self-referential behavioral metric**: model-based ρ and JS measure alignment with the model's own predictions. The EB metric partially addresses this, but its coverage (5 action categories, 6 game-state buckets, nine players in the EB substitute-quality protocol) is limited relative to the model's fine-grained conditional policies.
- **Pseudo-ground-truth provenance**: the probe-pair appendix mixes sources of uneven rigor, and several entries are supported only by broad public comparisons rather than exact expert-validated similarity labels.
- **No strong learned baselines**: comparisons against a sequential encoder or MLP over possession summaries would clarify the graph formulation's contribution.
- **Weight-dependent selection**: the composite ranking depends on manually chosen weights.
- **Temporal stability**: fixed data window; embeddings may not track evolving styles.
- **Low-data players**: the 50-possession minimum excludes many real scouting targets.
- **Retrieval scale**: galleries contain roughly 1,600 players; production use would require much larger galleries.
- **Counterfactual leakage**: actor–event edges allow actor identity to bleed into h^{event} before trait substitution.

11 Conclusion

We define player similarity as similarity of conditional decision policies and propose a pipeline that learns behaviorally grounded embeddings through heterogeneous graph encoding, attention-based trait extraction, and FiLM conditioning. Across 11 ablation variants and a broad evaluation suite, the paper supports four restrained conclusions:

1. **Position ablation plus split-context edges is beneficial in the tested setup.** Within the evaluated grid, this combination gives the strongest non-collapsed cross-context behavior.
2. **Player-aware sampling appears misaligned with substitute retrieval.** It improves self-re-identification more than cross-player functional similarity.
3. **Acts-in dropout is a strong ingredient in the best-performing family.** The EB protocol supports this family as producing better observed behavioral matches than the weaker learned alternatives shown here, though the protocol is not granular enough for clean causal isolation.
4. **The selected model is best interpreted as a weighted best-compromise model.** `acts_in_dropout_pos_gu` ranks first under the chosen composite, but it does not dominate every metric and should therefore be presented as the most balanced tested configuration, not the universally best model.

The paper's main methodological contribution is promising, but the empirical claims should remain calibrated to the strength of the current validation evidence.

Code Availability

The full code used for preprocessing, graph construction, model training, evaluation, and figure generation is available in the accompanying repository.

Appendix

A Detailed Evaluation Procedures

This appendix explains in detail what each evaluation measures, how it is conducted, and how it should be interpreted. The goal of the evaluation suite is not to rely on a single number, but to assess player embeddings from several complementary angles: (i) whether the model can still predict actions on held-out data, (ii) whether nearest neighbors agree with a small probe set of known comparisons, (iii) whether a player's embedding is stable across contexts, (iv) whether cosine proximity corresponds to behavioral similarity under controlled situations, (v) whether this remains true when behavior is measured from *observed* actions rather than the model's own predictions, and (vi) whether the retrieved neighbors are externally plausible to a coach or scout.

A.1 Overview of the evaluation suite

The main paper evaluates each GNN model on 14 metrics grouped into six categories: *model-based behavioral fidelity*, *empirical behavioral fidelity*, *cross-context robustness*, *pseudo-ground-truth retrieval*, *external validation*, and *supplementary predictive stability*. The purpose of this organization is to prevent over-interpreting any one metric. For example, a model may preserve within-player identity very well while still failing to retrieve functionally similar *other* players, or it may score well on a small probe set while behaving poorly under counterfactual action testing. Accordingly, the suite should be read as a structured set of checks rather than a single benchmark.

A.2 Held-out predictive validity

The first evaluation asks whether the model remains a competent conditional action predictor on a held-out test split of matches. This is a *proxy-task* evaluation rather than the final substitute-retrieval objective, but it is important because a degenerate model with poor action prediction is unlikely to yield meaningful player embeddings.

How it is conducted. After training, the best checkpoint is evaluated on the held-out test split using the same masked-input setting as in training. The model predicts: (i) the next action type (14-way classification), (ii) the action direction / angle bin (9-way classification), (iii) the action length bin (5-way classification), and (iv) two binary possession-level outcomes: whether the possession ends in a shot and whether it ends in a goal. The reported outputs include accuracy and macro-F1 for the multiclass heads, and accuracy / BCE / AUC-style diagnostics for the binary outcome heads.

What it measures. This evaluation measures whether the state pathway, trait pathway, and supervision heads remain behaviorally meaningful on unseen matches. A high score indicates that the representation still encodes useful predictive information about what actions are likely in a given masked situation.

How to interpret it. This metric should not be confused with substitute quality. A model can be a slightly weaker predictor yet yield a more useful embedding geometry for retrieval. In this project, held-out predictive validity is therefore treated as a supporting check and not as the primary selection criterion.

A.3 Pseudo-ground-truth retrieval

Pseudo-ground-truth retrieval tests whether the embedding space retrieves a small set of plausible, human-recognizable substitute relationships.

How it is conducted. A probe set of 15 directional player-similarity pairs is used. These pairs are divided into three tiers according to how strong the expected similarity is. For each pair (A, B), the system computes the cosine similarity between z_A and z_B, then finds the rank of B in A's nearest-neighbor list and the rank of A in B's nearest-neighbor list. Similarity-based modes apply gender filtering, so male queries are compared only against male candidates and female queries only against female candidates when the gender map is available.

The aggregate report includes:

- mean rank and median rank,
- Hit @5, Hit @10, Hit @20, and Hit @50,
- per-tier summaries,
- and bootstrap 95% confidence intervals for mean rank and Hit @10.

What it measures. This evaluation asks a simple question: if two players are widely regarded as similar, does the learned embedding space place them near each other?

How to interpret it. This metric is best viewed as a *directional sanity check.* The pair list is useful because it exposes obviously unreasonable neighbors, but it is not expert-validated ground truth. As a result, disagreement with this probe set should not automatically be interpreted as model failure. It is most informative when combined with the stronger behavioral and self-consistency evaluations below.

A.4 Cross-context self-consistency

Self-consistency tests whether a player's embedding remains stable when that same player is observed in different subsets of data.

A.4.1 Competition-split self-consistency

How it is conducted. For each player, possession-level player embeddings are first generated by running inference on the possession graphs and collecting the local player-node representation from every possession in which that player appears. Players who appear in at least two competitions with enough possessions in each split are retained. Their possession-level embeddings are then pooled *separately* for the two competitions using the same learned pooling operator as in normal embedding generation, producing two vectors:

$$z_p^{(a)}, \quad z_p^{(b)}.$$

The system then checks whether $z_p^{(a)}$ retrieves $z_p^{(b)}$ as its own nearest match, and vice versa.

Importantly, this is done as an *open-set* retrieval task. Players who are not testable under the two-competition criterion but still have enough total possessions are pooled once and inserted into the retrieval gallery as distractors. This makes the task harder and closer to the real inference-time search setting.

Reported metrics. The evaluation reports:

- self-cosine mean / median / standard deviation,
- cross-cosine mean against other players,
- cosine margin,
- self-retrieval mean rank and median rank,
- Hit @1, Hit @5, Hit @10, Hit @20, and Hit @50,
- and position-group breakdowns.

What it measures. This tests whether the embedding captures player identity and behavior robustly across competitions, leagues, or tournaments. It is therefore a cross-context robustness test.

A.4.2 Random-half self-consistency

How it is conducted. For players with at least 100 possessions, the possession set is randomly shuffled and split 50/50 into two halves. Each half is pooled independently to produce two vectors for the same player. Retrieval is then performed exactly as above, again with open-set distractors when possible.

What it measures. This tests whether the embedding is stable under a within-player resampling perturbation when there is no competition shift. It is an easier test than competition-split evaluation because the two halves usually come from a more homogeneous distribution.

How to interpret the two tests together. Random-half consistency checks *within-context* stability, whereas competition-split consistency checks *cross-context* stability. A model that performs well only on random-half retrieval may be good at clustering repeated samples of the same player without necessarily learning a representation that transfers across competitions. This distinction is important in this project because player-aware sampling strongly improves within-player retrieval while not necessarily improving substitute discovery.

A.5 Model-based behavioral diagnostics

The policy diagnostic evaluates whether the *geometry* of the learned embedding space matches the model's own notion of behavioral similarity.

A.5.1 Canonical-situation construction

The policy diagnostic begins by sampling a fixed set of *canonical situations*. These are real events drawn from the dataset and stratified by actor position group so that defenders, midfielders, forwards, and goalkeepers all contribute situations. For each sampled event, the graph is masked, encoded, and the corresponding event-state vector h^{event} is extracted.

For every player embedding z_p, the model then applies FiLM conditioning to that same state and produces predicted distributions for:

$$P_{\text{type}}(\cdot \mid s, p), \quad P_{\text{angle}}(\cdot \mid s, p), \quad P_{\text{length}}(\cdot \mid s, p).$$

Thus, each player is asked: *if you were placed in this exact same state, what would you do?*

A.5.2 Policy-distance correlation

How it is conducted. For each pair of players (p, q) in a valid comparison pool, the evaluation computes the Jensen–Shannon divergence between their predicted distributions in every sampled situation and averages across the three heads:

$$d_{\text{model}}(p, q) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \frac{1}{3} \sum_{h \in \{\text{type,angle,length}\}} \text{JS}\Big(P_h(\cdot \mid s, p), P_h(\cdot \mid s, q)\Big).$$

This is then compared against cosine distance

$$d_{\cos}(p, q) = 1 - \cos(z_p, z_q).$$

The reported summary is Spearman's ρ between $d_{\text{model}}(p, q)$ and $d_{\cos}(p, q)$, computed within valid player groups.

When gender metadata is available, the comparison is further restricted within gender. The correlation is computed within position-group (and optionally position-group×gender) subsets rather than across the entire player pool, because comparing goalkeepers to forwards or male to female players would create trivial large-distance pairs that are not relevant to the substitute-setting question.

Statistical testing. Significance is assessed with a Mantel-style row-permutation test rather than a naive parametric test. This matters because pairwise distances in a matrix are not i.i.d.: many pairs share players. Group-level p-values are then corrected with Holm–Bonferroni.

What it measures. This is the most direct test of whether embedding distance tracks *counterfactual behavioral distance* under the model. High ρ means that players who are nearby in cosine space also receive similar predicted action distributions in matched situations.

Interpretation. A high correlation is desirable, but it is still partly self-referential because both the embeddings and the action distributions come from the same trained model. For that reason, the empirical-behavioral evaluation in Section A.6 is essential as an external cross-check.

A.5.3 Absolute top-k behavioral distance

How it is conducted. For each query player, the $k = 10$ nearest neighbors in cosine space are identified within the allowed comparison pool. The model-based Jensen–Shannon distance from the query to each of those neighbors is computed using the same canonical situations and the same three output heads. The reported quantity is the mean of these distances:

$$\text{Top-}k \text{ JS} = \frac{1}{k}\sum_{j=1}^{k} d_{\text{model}}(p, n_j).$$

What it measures. This is an *absolute* behavioral-quality metric. It asks whether the nearest neighbors are actually behaviorally similar according to the model, not just whether they are better than random.

Why it matters. This quantity is especially important because ratio-based summaries can sometimes look strong simply because the random baseline is very poor. Absolute top-k JS directly measures the behavioral coherence of the retrieved neighbors themselves.

A.5.4 Substitute quality ratio

How it is conducted. A set of query players is selected using a mixed strategy: roughly half are high-possession players from each position group and the rest are random players from the remaining pool. For each query player, the evaluation forms:

- a top-k cosine-neighbor set, and
- a random-k baseline set drawn from the same position group (and same gender when applicable).

It then computes the mean model-based JS to the top-k set and to the random-k set. The aggregate ratio is

$$R_{\text{model-sub}} = \frac{\text{JS}_{\text{random}}}{\text{JS}_{\text{top-}k}}.$$

What it measures. If $R_{\text{model-sub}} > 1$, the retrieved neighbors are behaviorally closer than random same-group players. Larger values indicate a stronger retrieval advantage over chance.

Interpretation. This metric is useful, but it must be interpreted carefully. A large ratio can arise either because the top-k neighbors are genuinely excellent or because the embedding space is unusually spread out, making the random denominator large. For that reason, the ratio should always be read together with the absolute top-k JS.

A.5.5 FiLM sensitivity

How it is conducted. This diagnostic tests whether the trait pathway is actually load-bearing. For each player, the correct z_p is replaced by a randomly chosen same-group player's embedding, using a derangement so that no player keeps their own vector. The model then recomputes the predicted action distributions in the same canonical situations and compares them against the correct predictions using JS divergence.

What it measures. If substituting another player's trait vector barely changes the predictions, then the model may be mostly ignoring z_p and relying instead on the state pathway or residual actor information. If the divergence is substantial, the trait embedding is actively modulating behavior.

Interpretation. High FiLM sensitivity means that z_p is genuinely influencing the prediction head, which is crucial for a substitute-retrieval system based on trait–state decomposition. This diagnostic is especially relevant because the architecture includes actor–event edges, so one must verify that the model has not collapsed into an almost state-only or actor-identity-dominated predictor.

A.6 Empirical behavioral fidelity

Empirical behavioral fidelity is the key non-self-referential behavioral check in the suite. Whereas the previous section compares embeddings to the *model's own predicted policies*, this section compares embeddings to *observed player actions* from the raw event data.

A.6.1 Observed-action histogram construction

How it is conducted. Each player's observed events are bucketed by a coarse state definition:

$$\text{pitch third} \times \texttt{under\_pressure},$$

which yields 6 buckets in total. Within each bucket, event types are collapsed into five coarse action categories:

$$\{\text{Pass}, \ \text{Carry}, \ \text{Shot}, \ \text{Dribble}, \ \text{Other}\}.$$

For a bucket to be valid for a given player, it must contain at least 15 events. Two players are compared only if they share at least 3 valid buckets.

For a shared bucket b, let $\hat{P}_b^{(p)}$ and $\hat{P}_b^{(q)}$ denote the empirical action histograms of players p and q. The empirical distance is then the average bucket-wise Jensen–Shannon divergence over shared buckets.

What it measures. This tests whether players that are nearby in the embedding space also behave similarly in *real observed decisions* under coarse matched situations.

Strengths and limitations. Its main strength is that it is not self-referential. Its weakness is that it is much coarser than the learned policy representation: only 6 state buckets and 5 action categories are used, and players must have enough observations in matching buckets. Thus, it is noisier but more externally grounded.

A.6.2 Empirical policy-distance correlation

How it is conducted. For valid player pairs within a comparison group, the system computes:

- cosine distance between global embeddings, and
- empirical JS divergence between observed-action histograms.

Spearman's ρ is then computed within group. If gender metadata is available, the analysis is split within gender as well. Pairs with too few shared valid buckets are skipped.

What it measures. This asks whether the embedding geometry agrees with observed, data-derived behavioral similarity rather than with the model's internal policy surrogate.

A.6.3 Empirical substitute-quality ratio

How it is conducted. A small set of query players with valid empirical histograms is chosen. For each, the evaluation compares the empirical JS to top-k cosine neighbors and to k random same-group players:

$$R_{\mathrm{EB}} = \frac{\mathrm{EmpJS}_{\mathrm{random}}}{\mathrm{EmpJS}_{\text{top-}k}}.$$

A bootstrap confidence interval is reported when enough query players are available.

What it measures. This directly tests whether embedding neighbors are *more behaviorally similar in real observed actions* than random same-group peers.

Interpretation. Because this metric uses real decisions rather than the model's own output head, it is one of the strongest pieces of evidence that the retrieved substitutes are behaviorally meaningful. At the same time, it is based on a coarse histogram construction and should therefore be treated as a conservative external check rather than a full replacement for the model-based diagnostics.

A.7 External validation

External validation asks whether retrieved substitutes look plausible under signals that were *not* used to train the model.

A.7.1 FIFA / EA FC attribute comparison

How it is conducted. The system first matches StatsBomb players to FIFA / EA FC records. It then samples query players by gender and position group (10 male and 10 female by design when available), with famous players added when possible. For each query player, the model retrieves the top-1 same-gender substitute among players who also have FIFA data.

The comparison is then performed in the FIFA attribute space. For outfield players, the main six attributes are compared:

$$\text{Pace},\ \text{Shooting},\ \text{Passing},\ \text{Dribbling},\ \text{Defending},\ \text{Physical}.$$

For goalkeepers, the goalkeeper-specific axes are used instead. The outputs include radar charts, summary plots, mean attribute differences, and position-group agreement statistics.

What it measures. This is an out-of-distribution sanity check. The GNN never sees FIFA ratings during training, so agreement in this space suggests that the learned retrieval is not completely divorced from a familiar human scouting representation.

How to interpret it. FIFA agreement is supportive but not definitive. FIFA ratings are subjective, commercially shaped, and do not directly measure conditional decision policy. The evaluation is also relatively small. For these reasons, FIFA comparison is best read as external corroboration rather than as a gold-standard target.

A.7.2 Head-coach qualitative scoring

How it is conducted. Six famous-player queries are selected: Neuer, Van Dijk, Alexander-Arnold, De Bruyne, Messi, and Mbappé. For each selected model, the top-5 nearest neighbors for each query are shown to an evaluator together with the query player's name, position, and broad style description. The evaluator then assigns a score from 1 to 5 based on whether the neighbor list looks like a plausible stylistic substitute set. In the present report, the evaluator is Claude Opus 4.6.

What it measures. This is a *face-validity* check. It asks whether the retrieved neighbors look reasonable to a football-aware reader in concrete, well-known examples.

How to interpret it. This evaluation is intentionally heuristic. It is not expert consensus, and inter-rater reliability has not been established. Its role is to catch glaringly implausible retrieval behavior and to supplement the quantitative metrics with domain-facing examples.

A.8 Supportive qualitative and counterfactual checks

A.8.1 Qualitative neighbor inspection

Beyond the scored head-coach tables, the report also inspects nearest-neighbor lists for notable players such as Cristiano Ronaldo, Toni Kroos, Messi, and others. These tables are not standalone benchmark metrics, but they are extremely useful for understanding *what kind* of errors a model makes. For example, some models return players who are nearby by nominal position, whereas others return players who are closer in creative function or decision profile.

A.8.2 Situation-level counterfactual analysis

The final analysis stage provides a more mechanistic validation. For each query player:

1. the top-5 nearest neighbors are retrieved,
2. five real game situations are sampled from the query player's possessions,
3. the masked state representation h^{event} is extracted for each situation,
4. different players' global trait vectors z_p are substituted into that same state,
5. and the predicted action distributions are compared side by side.

What it measures. This directly tests the motivating claim of the project: *if these players were placed in the same situation, would they choose similar actions?*

Important caveat. Because actor–event edges are present at inference time, some actor identity can leak into the event representation before trait substitution. Accordingly, this analysis should be interpreted as evidence of *FiLM sensitivity under realistic inference conditions*, not as a perfectly clean intervention on a purely identity-free state representation.

A.9 Why multiple evaluations are necessary

No single evaluation in this paper is sufficient on its own.

Why test metrics are insufficient. A model can predict actions well but still produce embeddings that are poor for retrieval.

Why pseudo-ground-truth retrieval is insufficient. The pair list is small and heuristic, and some pairs are based on public narratives rather than expert tactical validation.

Why model-based behavioral metrics are insufficient. They are informative, but they partially compare the embedding space to the model's own policy surrogate.

Why empirical behavioral fidelity is insufficient. It is externally grounded, but it is much coarser than the full conditional policy and can miss fine-grained tactical distinctions.

Why qualitative checks are insufficient. They are useful for face validity but are inherently subjective.

Practical reading of the suite. The strongest overall evidence appears when several views agree:

- a model remains non-degenerate on held-out action prediction,
- retrieves plausible pseudo-ground-truth neighbors,
- preserves player identity across data splits,
- aligns cosine distance with policy distance,
- also aligns cosine distance with observed-action similarity,
- and produces externally reasonable substitutes under FIFA and qualitative inspection.

This is why the paper uses a broad evaluation suite and a weighted composite, rather than selecting a model from any single table.

A.10 Summary of what each evaluation answers

For clarity, the evaluations can be summarized as follows:

Evaluation	**Core question answered**
Held-out predictive validity	Can the model still predict masked actions and outcomes on unseen matches?
Pseudo-ground-truth retrieval	Do widely plausible substitute pairs land near each other in the embedding space?
Competition-split self-consistency	Is the same player stably represented across different competitions?
Random-half self-consistency	Is the same player stably represented under within-player resampling?
Policy-distance correlation	Does cosine distance track counterfactual behavioral distance under the model?
Absolute top-k JS	Are the retrieved neighbors themselves behaviorally close in absolute terms?
Substitute quality ratio	Are retrieved neighbors better than random same-group players?
FiLM sensitivity	Does the trait vector z_p actually affect the model's predictions?
Empirical behavioral correlation	Does cosine distance track similarity in *observed* actions?
Empirical substitute ratio	Are retrieved neighbors better than random under *observed*-action behavior?
FIFA comparison	Do retrieved substitutes look externally plausible in a human-curated attribute space?
Qualitative head-coach scoring	Do the neighbor lists pass a realistic football face-validity check?
Situation-level counterfactual analysis	In explicit real situations, do query players and retrieved candidates make similar predicted decisions?

B Pseudo Ground-Truth Pairs

The 15 pairs below form a *heuristic probe set* rather than a formal ground-truth benchmark. They were assembled from a mixture of public analytics articles, recruitment pieces, public similarity tools, and analyst-style media comparisons. The purpose of this set is only to test whether the learned space retrieves obviously unreasonable neighbors or broadly plausible stylistic alternatives.

Because the underlying sources are heterogeneous and not all are equally rigorous, disagreement with this table should not automatically be interpreted as model failure. For that reason, the main paper treats pseudo-GT results as directional sanity checks rather than definitive retrieval supervision.

Selection criteria. The probe pairs were chosen to span multiple position groups, include both men's and women's football, and reflect commonly discussed stylistic comparisons in public discourse. They are intentionally heterogeneous and are therefore not suitable for fine-grained statistical claims. Their value is diagnostic: they help identify retrieval outputs that are clearly implausible, but they do not provide a gold-standard ranking target.

Table 7: Pseudo ground-truth similar-player pairs.

Tier	Player A	Player B	Positions	Poss A	Poss B	Heuristic Rationale
1	Modrić	Kroos	RCM / LDM	828	584	Tempo-controlling deep playmakers (Real Madrid partnership).
1	Van Dijk	R. Dias	LCB / LCB	562	618	Dominant ball-playing centre-backs; common public comparison.
1	J. Alba	Robertson	LB / LWB	570	255	Widely discussed attacking fullback comparison from recruitment-style analysis.
1	TAA	Hakimi	RDM / RB	87	333	Attack-first right-sided creators; included despite TAA's low possession count.
1	Bonmatí	Putellas	RCM / LCM	854	465	High-level women's midfield comparison with overlapping creative profiles.
2	Saka	Dembélé	RW / RW	518	407	Inverted-winger archetype comparison.
2	Kane	Lewandowski	CF / LCF	676	335	Clinical #9s with strong finishing and link play.
2	Wirtz	De Bruyne	LAM / CAM	1820	487	Publicly framed as advanced creators with similar attacking influence.
2	Renard	Bright	LCB / RCB	479	639	Widely recognized top-level women's centre-back comparison.
2	Hemp	Caldentey	LW / LW	787	840	Creative left-sided forwards with hybrid scoring and chance-creation profiles.
2	Kroos	Enzo F.	LDM / CDM	584	351	Deep midfield replacement-style comparison.
2	Bellingham	Griezmann	CAM / CAM	607	690	Goal-threat attacking-midfield comparison; noisier than tier-1 pairs.
3	Musiala	Foden	LW / LW	442	477	Common stylistic comparison, but role nuance differs.
3	Yamal	N. Williams	RW / LW	252	317	Public comparison tied partly to shared tournament context rather than same-side role.
3	Neuer	Donnarumma	GK / GK	339	296	Distribution-oriented elite goalkeepers; stylistic details still differ.

C Full Raw Metrics and Weights

Table 8: Evaluation metrics with category weights.

#	Metric	Wt	Category (total)
1	Spearman ρ (model-based)	.14	Model-Based Beh. (0.28)
2	Absolute Top-k JS (model-based)	.10	
3	Sub Quality Ratio (model-based)	.04	
4	EB Spearman ρ	.07	Empirical Beh. (0.09)
5	EB Substitute Ratio	.02	
6	Comp-Split Mean Rank	.17	Cross-Context (0.23)
7	Comp-Split Hit@10	.06	
8	Random-Half Mean Rank	.05	Supplementary (0.08)
9	Test F1 Avg	.03	
10	Pseudo-GT Mean Rank	.08	Pseudo-GT (0.12)
11	Pseudo-GT Hit@50	.04	
12	FIFA Main-6 Diff	.05	External (0.20)
13	FIFA Position Match%	.05	
14	Qualitative Avg	.10	

Table 9: Raw metric values for all 11 GNN models. Direction: $\rho\uparrow$, JS $\downarrow$, MR $\downarrow$, H@K $\uparrow$, Ratio $\uparrow$, FIFA $\downarrow$, Pos% $\uparrow$, F1 $\uparrow$, Qual $\uparrow$, EB $\rho\uparrow$, EB Ratio $\uparrow$. "—" = unavailable.

Model	ρ	JS	Comp MR	C.H@10	Rnd MR	GT MR	GT H@50	Ratio	FIFA	Pos%	F1	Qual	EB ρ	EB Rat	**Score**
acts_in_dropout_pos_gu	.922	.00105	74.0	.328	8.2	66.7	.567	7.62	7.7	50	.660	3.5	.195	1.75	**.867**
acts_in_dropout	.928	.00103	68.9	.351	8.7	68.2	.633	7.75	7.9	50	.657	3.0	.207	1.95	.864
pos_abl_split_ctx_ema_v2	.934	.00145	67.5	.349	6.1	64.3	.633	6.35	8.8	50	.674	—	—	—	.860†
pos_ablated	.974	.00111	81.3	.275	13.9	69.6	.567	8.59	8.6	50	.687	—	.196	1.66	.847†
pos_abl_split_ctx_ema	.944	.00104	67.7	.318	9.3	62.6	.700	7.95	9.1	56	.650	3.0	.207	1.72	.846
pos_abl_split_ctx	.929	.00141	67.3	.346	6.3	63.1	.633	6.63	9.0	50	.675	2.8	.207	1.62	.820
player_samp	.788	.00112	93.5	.264	2.9	70.2	.333	11.91	8.9	50	.700	—	.223	1.56	.735†
split_ctx_ps	.778	.00103	89.3	.262	2.7	71.4	.333	13.11	9.2	25	.688	—	.230	1.70	.692†
baseline	.744	.00330	81.6	.238	26.1	67.6	.667	3.84	8.6	44	.692	2.2	.188	1.32	.593
split_ctx	.710	.00461	83.9	.215	36.4	56.1	.600	3.13	8.5	56	.684	1.8	.189	1.25	.536
pos_abl_split_ctx_v2	.686	.00696	200.8	.076	177.8	198.8	.200	3.07	9.8	38	.639	—	.104	1.21	.023†

† Scored on fewer than 14 metrics; weight renormalized over available dimensions.

D Per-Query Qualitative Scoring

Six famous-player queries were scored by an LLM evaluator (Claude Opus 4.6) on a 1–5 scale for face-validity of top-5 neighbors. The evaluator was given the query player's name, position, and broad playing style, together with the top-5 retrieved neighbors and cosine similarities, and asked whether the neighbors were stylistically plausible substitutes (5 = excellent match, 1 = poor or wrong position group). These scores are heuristic and directional, not authoritative; inter-rater reliability with human experts has not been assessed.

Table 10: Per-query qualitative scores (1–5). M1=baseline, M2=`pos_abl_split_ctx`, M3=`split_ctx`, M4=`pos_abl_split_ctx_ema`, M5=`acts_in_dropout`, M6=`acts_in_dropout_pos_gu`.

Query	M1	M2	M3	M4	M5	M6
Neuer (GK)	2	3	3	**4**	3	**4**
Van Dijk (CB)	3	2	3	2	3	**4**
TAA (RDM)	**3**	2	1	2	2	2
De Bruyne (CAM)	3	3	2	3	3	**4**
Messi (RCF)	1	**3**	1	**3**	**3**	**3**
Mbappé (LW)	1	**4**	1	**4**	**4**	**4**
Total	13	17	11	18	18	**21**

Key patterns. Position-ablated models (M2, M4–M6) outperform the more position-driven models (M1, M3) on creative or unusual players such as Messi and Mbappé. `split_ctx` tends to retrieve positional peers rather than stylistic matches. The selected model retrieves Rúben Dias for Van Dijk and Bruno Fernandes for De Bruyne, which helps explain its higher qualitative average. No model scores 5 on any query, and TAA remains difficult across all variants.

E Model Variation Details

Each pipeline variant is identified by a shorthand name that concatenates its active components. Table 11 provides the full specification.

Table 11: Complete hyperparameter specification for every pipeline variant. Defaults (applied when a cell is blank): split-context = off, position ablation = off, player sampling = off, t=2.0, λ_{uni}=0.3, EMA = off, acts-in dropout p=0, λ_{pos}=0, group weight w_g=1.

Name	Split Ctx	Pos Abl	Plyr Samp	t	λ_{uni}	EMA	λ_{align}	Drop p	λ_{pos}	w_g	Motivation
`baseline`				2	.3					1	Default configuration; reference point.
`player_samp`			✓	2	.5					1	Test whether structuring batches by player helps.
`split_ctx`	✓			2	.3					1	Separate teammate/opponent context edges.
`split_ctx_ps`	✓		✓	2	.5					1	Combine split-context with player sampling.
`pos_ablated`		✓		2	.3					1	Remove position from FiLM input.
`pos_abl_split_ctx`	✓	✓		4	1.0					1	Ablation + split-context + strong uniformity.
`pos_abl_split_ctx_v2`	✓	✓		4	0.7					1	Reduced uniformity (λ=0.7); stress-test for collapse sensitivity.
`pos_abl_split_ctx_ema`	✓	✓		4	1.0	✓	.3			1	Add EMA cross-batch alignment.
`pos_abl_split_ctx_ema_v2`	✓	✓		4	1.0	✓	.1			1	Lower alignment weight; archived comparison variant.
`acts_in_dropout`	✓	✓		4	1.0			.3		1	Drop actor edges during training to force stronger reliance on the trait pathway.
`acts_in_dropout_pos_gu`	✓	✓		4	1.0			.3	.3	3	Add position-group auxiliary loss and stronger within-group repulsion for improved qualitative coherence.

F FIFA Radar Diagram

Figure 4 provides an external sanity check by comparing the FIFA/EA FC attribute profiles of each query player and the model's top-1 retrieved substitute. The six outfield attributes (Pace, Shooting, Passing, Dribbling, Defending, Physical) are composite ratings on a 0–100 scale compiled by EA Sports from scouting networks; goalkeeper pairs instead use Diving, Handling, Kicking, Positioning, and Reflexes. Because the GNN is trained exclusively on StatsBomb on-ball decisions and never sees FIFA data, agreement between the radar profiles is a genuine out-of-distribution validation signal. Across the 16 non-GK pairs evaluated for ACTS_IN_DROPOUT_POS_GU, the mean main-6 absolute difference is 7.7 (Table 5), substantially closer than a random same-position baseline but still above the FIFA heuristic (5.0), which optimises directly on that attribute space.

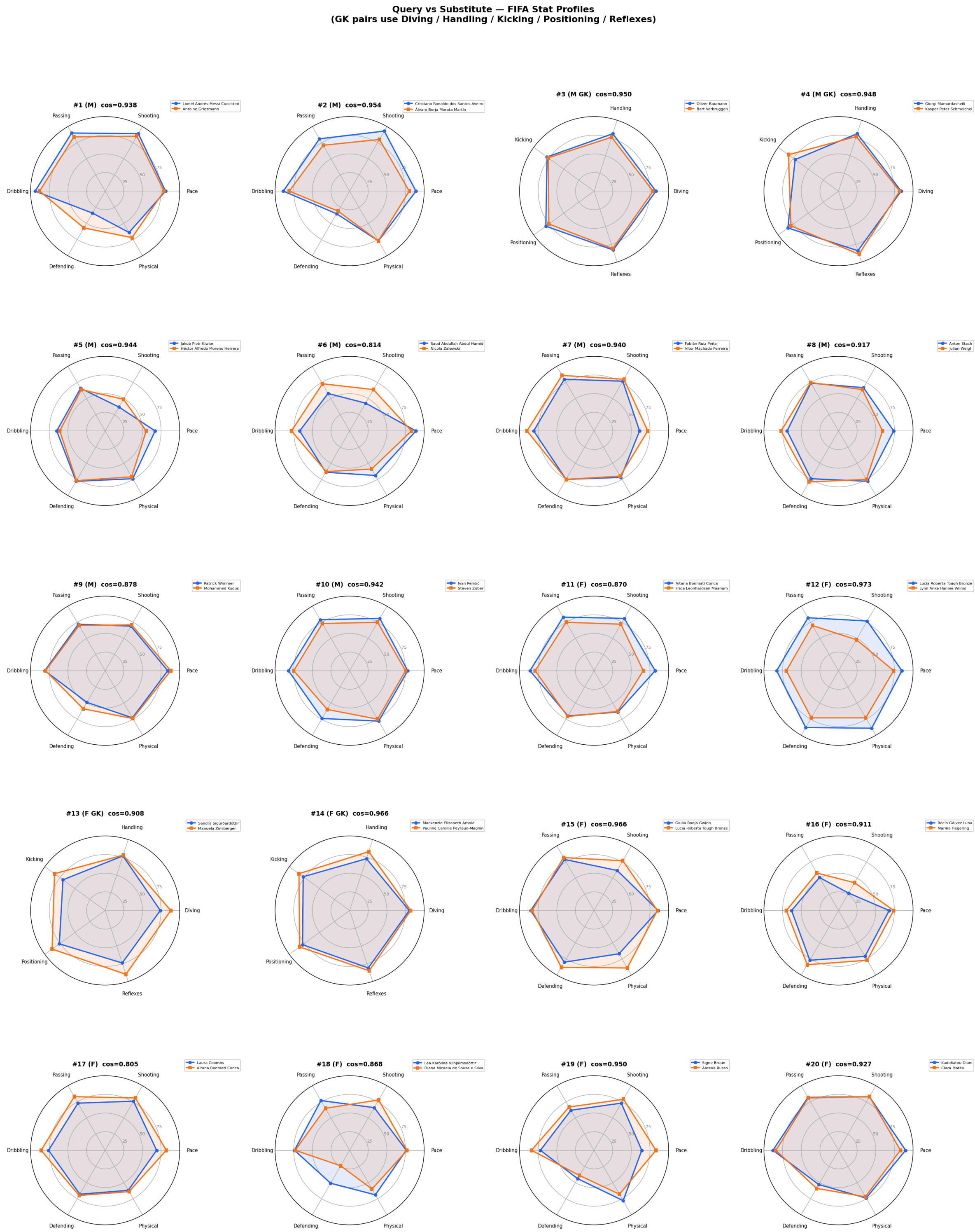


Figure 4: FIFA attribute radar profiles for query–substitute pairs (ACTS_IN_DROPOUT_POS_GU). Blue = query, orange = top-1 cosine substitute. Outfield pairs use six main attributes; goalkeeper pairs use five GK-specific axes. Close polygon overlap indicates agreement between GNN behavioral similarity and external FIFA attributes.

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